

Autodesk® Scaleform®

AMP

Scaleform

(AMP)

□

Alex Mantzaris

3.03

2013 4 10

Copyright Notice

Autodesk® Scaleform® 4.4

© 2014 Autodesk, Inc. All rights reserved. Except as otherwise permitted by Autodesk, Inc., this publication, or parts thereof, may not be reproduced in any form, by any method, for any purpose.

Certain materials included in this publication are reprinted with the permission of the copyright holder.

The following are registered trademarks or trademarks of Autodesk, Inc., and/or its subsidiaries and/or affiliates in the USA and other countries: 123D, 3ds Max, Algor, Alias, AliasStudio, ATC, AutoCAD LT, AutoCAD, Autodesk, the Autodesk logo, Autodesk 123D, Autodesk CAM 360, Autodesk Homestyler, Autodesk Inventor, Autodesk MapGuide, Autodesk Streamline, AutoLISP, AutoSketch, AutoSnap, AutoTrack, Backburner, Backdraft, Beast, BIM 360, Burn, Buzzsaw, CADmep, CAiCE, CAMduct, CFdesign, Civil 3D, Cleaner, Combustion, Communication Specification, Configurator 360™, Constructware, Content Explorer, Creative Bridge, Dancing Baby (image), DesignCenter, DesignKids, DesignStudio, Discreet, DWF, DWG, DWG (design/logo), DWG Extreme, DWG TrueConvert, DWG TrueView, DWGX, DXF, Ecotect, ESTmep, Evolver, FABmep, Face Robot, FBX, Fempro, Fire, Flame, Flare, Flint, FMDesktop, ForceEffect, FormIt, Freewheel, Fusion 360, Glue, Green Building Studio, Heidi, Homestyler, HumanIK, i-drop, ImageModeler, Incinerator, Inferno, InfraWorks, InfraWorks 360, Instructables, Instructables (stylized robot design/logo), Inventor, Inventor HSM, Inventor LT, Kynapse, Kynogon, LandXplorer, Lustre, MatchMover, Maya, Maya LT, Mechanical Desktop, MIMI, Mockup 360, Moldflow Plastics Advisers, Moldflow Plastics Insight, Moldflow, Moondust, MotionBuilder, Movimento, MPA (design/logo), MPA, MPI (design/logo), MPX (design/logo), MPX, Mudbox, Navisworks, ObjectARX, ObjectDBX, Opticore, Pipeplus, Pixlr, Pixlr-o-matic, Productstream, Publisher 360, RasterDWG, RealDWG, ReCap, ReCap 360, Remote, Revit LT, Revit, RiverCAD, Robot, Scaleform, Showcase, Showcase 360 ShowMotion, Sim 360, SketchBook, Smoke, Socialcam, Softimage, Sparks, SteeringWheels, Stitcher, Stone, StormNET, TinkerBox, ToolClip, Topobase, Toxik, TrustedDWG, T-Splines, ViewCube, Visual LISP, Visual, VRED, Wire, Wiretap, WiretapCentral, XSI.

All other brand names, product names or trademarks belong to their respective holders.

Disclaimer

THIS PUBLICATION AND THE INFORMATION CONTAINED HEREIN IS MADE AVAILABLE BY AUTODESK, INC. "AS IS." AUTODESK, INC. DISCLAIMS ALL WARRANTIES, EITHER EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO ANY IMPLIED WARRANTIES OF MERCHANTABILITY OR FITNESS FOR A PARTICULAR PURPOSE REGARDING THESE MATERIALS.

Autodesk Scaleform

AMP

Autodesk Scaleform Corporation

6305 Ivy Lane, Suite 310

Greenbelt, MD 20770, USA

www.scaleform.com

info@scaleform.com

(301) 446-3200

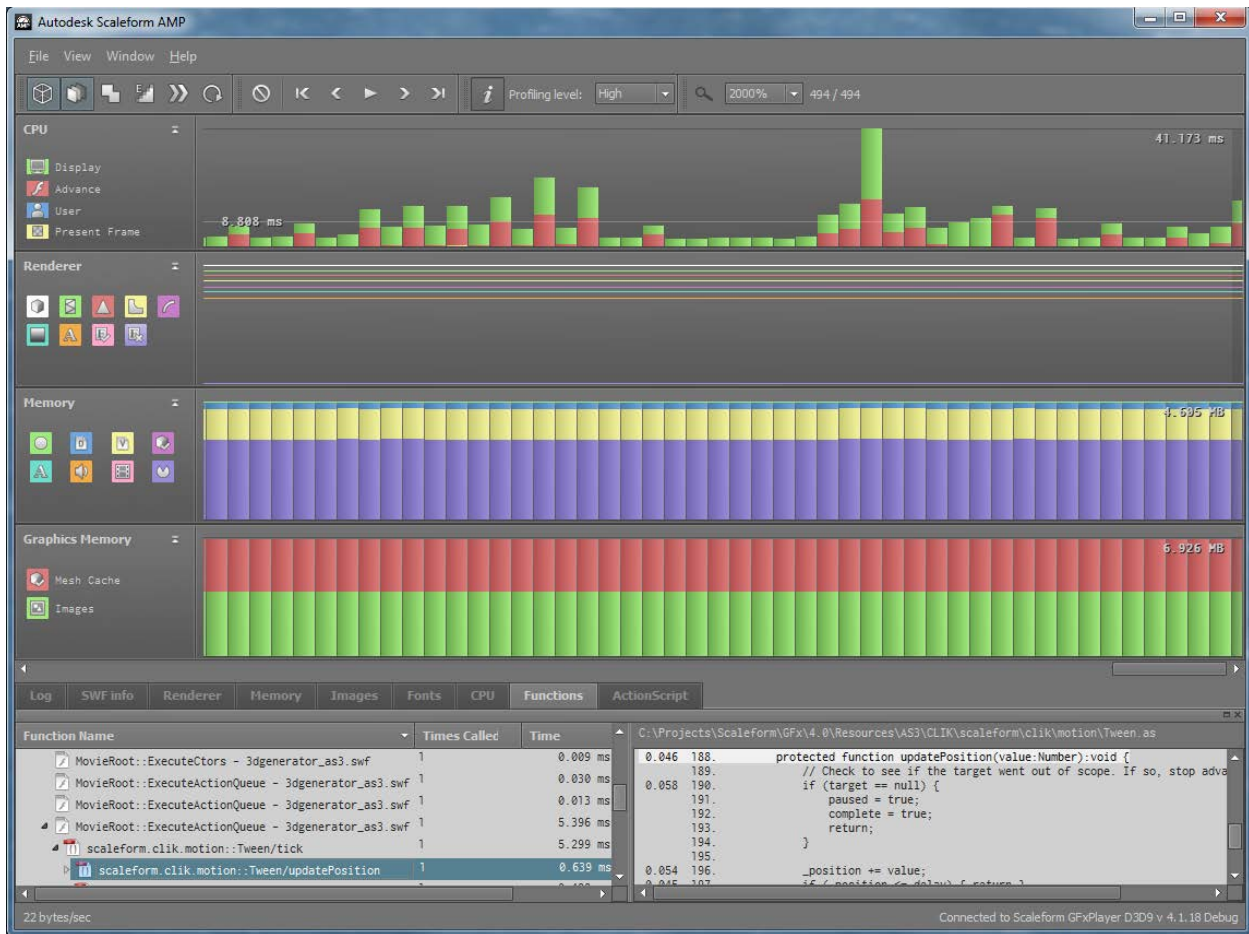
(301) 446-3199

1		1
1.1	AMP	1
1.2	AMP	2
2	AMP	10
2.1		10
2.2		11
2.3		12
2.4		13
2.5		19
2.6		19
3		21
4		22
5	AMP	23
5.1	Debug	23
5.2		23
5.3		24
5.4		24
6	AMP	25
7		26

1

AMPTM Scaleform® , CPU Scaleform

-
- AMP ActionScript , □



1 Scaleform AMP

Scaleform AMP , AMP , AMP , AMP

AMP Scaleform 3.2 □

1.1 AMP

Scaleform (AMP) □
ActionScript C++ Scaleform □ ActionScript 2

, ActionScript (ActionScript 3
)
 ActionScript Flash
 AMP AMP
 Scaleform Shipping (Builds) AMP
 AMP
 Movie AMP ActionScript SWF
 AMP Movies
 ,AMP
 Flash (Anti-
 Aliasing)
 AMP
 Scaleform Player
 Scaleform
 ,AMP
 (),
 ,AMP Flash (ActionScript 2 only)
 (ActionScript2) AS (ActionScript3) AMP SWD

1.2 AMP

(AMP) AMP (Profile)
 Scaleform Scaleform CLIK
 AMP Flash :

- o CPU (User) ActionScript (Advance) Scaleform (Display) Direct Access API Scaleform
- o CPU CPU , ,
- o :Scaleform , ,
- o : Scaleform (,
- o :Scaleform AMP , AMP
- o GPU
- o “ ” : “ ” (Log) ,
- o 性能顧問(Performance Adviser) : Flash
- o : (Anti-alias for readability) ,
- o : (Anti-alias for animation),
- o : (1.0) ,
- o :
- o :
- o :
- o :
- o , : ,
- o : (Best Practices) ,Flash
- o :
- o :
- o :
- o DirectAccess API :Direct Access API
- o :
- o :

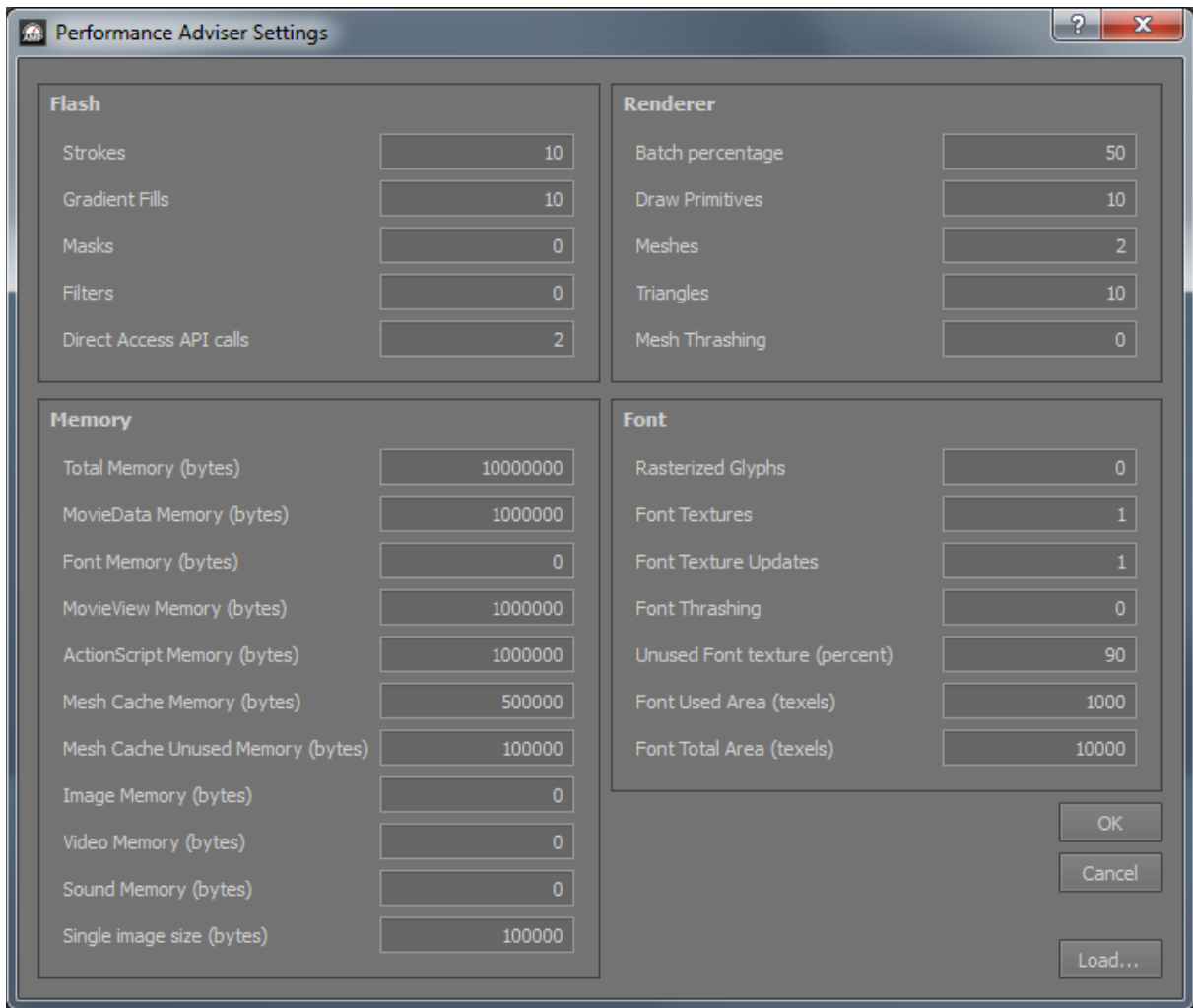
```

0      :
0      :
      ,      ,      ,
0      :
0      :
0      :
0      :Scaleform
0 MovieData      :
0      :
0 MovieView      :      Gfx::Movie      ,      MovieView

0 ActionScript      :ActionScript
0      :      ,
0      :      ,
0      :
0      :      Scaleform      ,      GPU
0      :
0      :      ,      AMP

      ,AMP      □      XML      ,      adviser.xml □
AMP      ,      □      □      (Performance Advisor)
□      (Settings)      ,      □

```

□ (OK), XML , AMP □

□ □

–adviser <AdviserFilePath> AMP,

</AdviserFilePath>□

o “SWF ” (SWF info) : flash

flash

o “ ” (Renderer) : , () (

, : () ,

)

o “ ” (Memory) : Scaleform Scaleform

(

AMP “i”) ,

- o “ ” (Images) :
 - AMP
 - Scaleform
 - ,DXT
- o “ ” (Fonts) : SWF “ ”
- o CPU : “ ” (Advance) “ ” (Display) “ ” (User)
- o “ ” (Functions) :
 - ActionScript C++
 - ActionScript VM
- o ActionScript : “ ” (Functions)
 - C++
- o **(Source)** :
 - ActionScript AS3,
 - AS SWF AS3
 - AS2
 - “(high) AS2
 - Scaleform
 - AMP SWF AS3 Flash AMP
 - SWF
- o **(File)**
 - o **(Connection)**
 - (Debug Info Paths):
 - ,AMP SWD (AS2) AS (AS3) “ ” (Add)
 - “ ” (Remove)
 - o **(Load Profile Frames)**

- o **(Save Profile Frames)** ☐
- o **(Dump Memory Report):** AMP ☐
- o **(Exit)** ☐
- o **(View) :**
 - o **(Graph Tooltips):** AMP ☐
 - o **(Time Units):** , “ ” ☐
 - o **(Memory Units):** “ ” “ ” ☐
- o **(Window) :** (SWF
 - o CPU
 - o ActionScript
 - o Restore UI(
- o **(Help)** AMP
 - o **(User Guide) --** Scaleform.com
 - o **(About) --** AMP
- o **UI** ☐
 - o  **(Wireframe):** ☐
 - o  : AMP (☐
 - o  ☐
 - o  **(Anti-aliasing):**
 - o Scaleform (EdgeAA) (HW FSAA) ☐
 - o (None) ☐

- 0  **(Fast-forward):** 快速前進 (Fast-forward) 功能允許用戶快速前進影片，以查看後續內容。此功能通常用於跳過影片中的冗餘部分，以節省時間。Flash 播放器支援此功能，並可透過影片控制欄中的快進按鈕來啟用。此功能可透過影片控制欄中的快進按鈕來啟用。此功能可透過影片控制欄中的快進按鈕來啟用。

0  **(Restart):** 重新開始 (Restart) 功能允許用戶將影片重新開始播放。此功能通常用於重新觀看影片，以確保用戶理解影片內容。Flash 播放器支援此功能，並可透過影片控制欄中的重新開始按鈕來啟用。此功能可透過影片控制欄中的重新開始按鈕來啟用。

0 “ ” (Frame control) : 影片控制 (Frame control) 功能允許用戶對影片進行更精確的控制。此功能通常用於查看影片中的特定幀，以進行詳細分析。Flash 播放器支援此功能，並可透過影片控制欄中的影片控制按鈕來啟用。此功能可透過影片控制欄中的影片控制按鈕來啟用。

0  **(Clear):** 清除 (Clear) 功能允許用戶清除影片中的元數據。此功能通常用於清除影片中的元數據，以確保影片內容的準確性。Flash 播放器支援此功能，並可透過影片控制欄中的清除按鈕來啟用。此功能可透過影片控制欄中的清除按鈕來啟用。

0  **(First Frame):** 第一幀 (First Frame) 功能允許用戶將影片播放頭移動到影片的第一幀。此功能通常用於查看影片的開頭部分，以確保影片內容的準確性。Flash 播放器支援此功能，並可透過影片控制欄中的第一幀按鈕來啟用。此功能可透過影片控制欄中的第一幀按鈕來啟用。

0  **(Previous Frame):** 上一幀 (Previous Frame) 功能允許用戶將影片播放頭移動到影片的上一个幀。此功能通常用於查看影片中的特定幀，以進行詳細分析。Flash 播放器支援此功能，並可透過影片控制欄中的上一幀按鈕來啟用。此功能可透過影片控制欄中的上一幀按鈕來啟用。

0  **(Pause):** 暫停 (Pause) 功能允許用戶暫停影片的播放。此功能通常用於暫停影片，以進行詳細分析。Flash 播放器支援此功能，並可透過影片控制欄中的暫停按鈕來啟用。此功能可透過影片控制欄中的暫停按鈕來啟用。

0  **(Next Frame):** 下一幀 (Next Frame) 功能允許用戶將影片播放頭移動到影片的下一個幀。此功能通常用於查看影片中的特定幀，以進行詳細分析。Flash 播放器支援此功能，並可透過影片控制欄中的下一幀按鈕來啟用。此功能可透過影片控制欄中的下一幀按鈕來啟用。

0  **(Last Frame):** 最後幀 (Last Frame) 功能允許用戶將影片播放頭移動到影片的最後幀。此功能通常用於查看影片的結尾部分，以確保影片內容的準確性。Flash 播放器支援此功能，並可透過影片控制欄中的最後幀按鈕來啟用。此功能可透過影片控制欄中的最後幀按鈕來啟用。
 - “ ” (Profiling) : 影片分析 (Profiling) 功能允許用戶對影片進行分析，以查看影片中的元數據。此功能通常用於查看影片中的元數據，以確保影片內容的準確性。Flash 播放器支援此功能，並可透過影片控制欄中的影片分析按鈕來啟用。此功能可透過影片控制欄中的影片分析按鈕來啟用。

0  **(Detailed Profiling):** 詳細影片分析 (Detailed Profiling) 功能允許用戶對影片進行更精確的分析，以查看影片中的元數據。此功能通常用於查看影片中的元數據，以確保影片內容的準確性。Flash 播放器支援此功能，並可透過影片控制欄中的詳細影片分析按鈕來啟用。此功能可透過影片控制欄中的詳細影片分析按鈕來啟用。

0  **(Zoom):** 放大 (Zooming out) 功能允許用戶放大影片，以查看影片中的細節。此功能通常用於查看影片中的細節，以確保影片內容的準確性。Flash 播放器支援此功能，並可透過影片控制欄中的放大按鈕來啟用。此功能可透過影片控制欄中的放大按鈕來啟用。

AMP

□

□

□

,

□

UI

AMP

,

“

”

“

UI” (Reset UI)

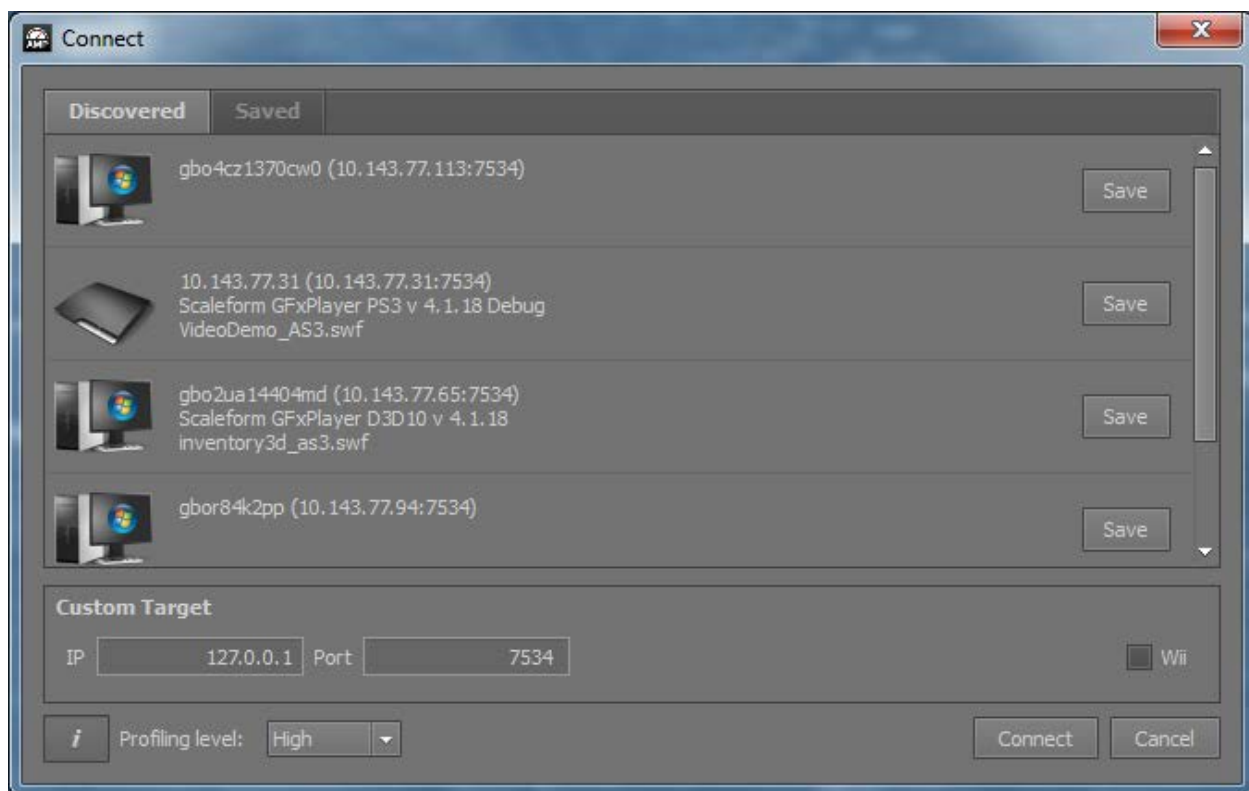
□

2 AMP

AMP , Scaleform Bin/AMP GFAmpClient.exe □

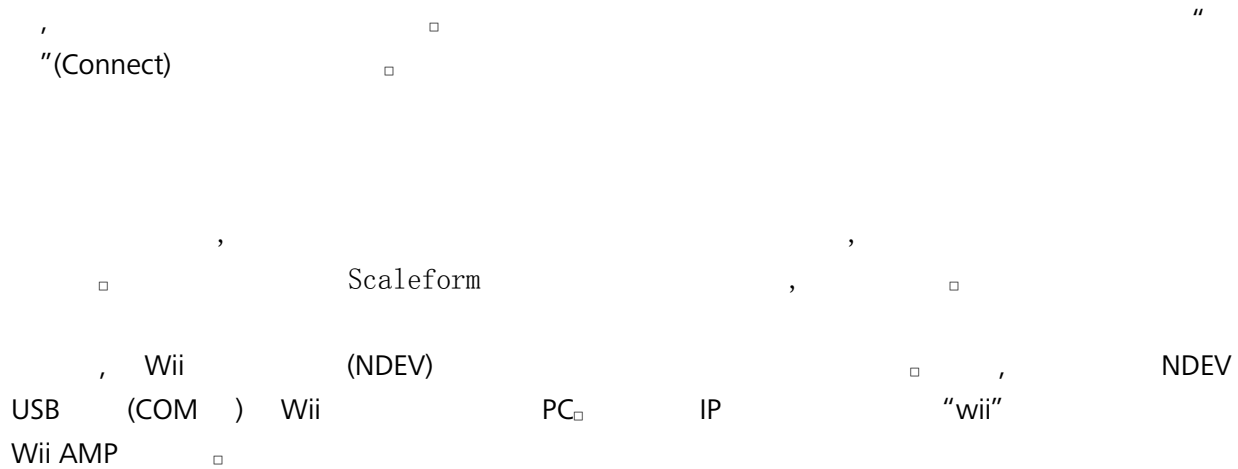
2.1

AMP
IP □ AMP , ,
□

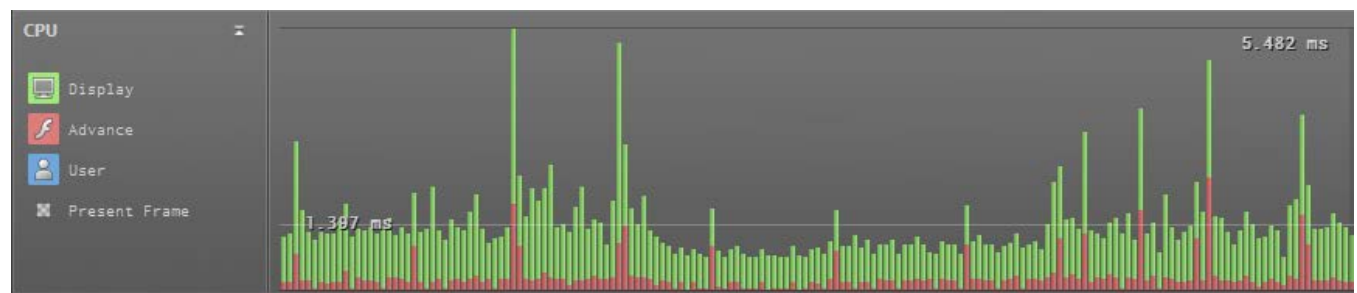
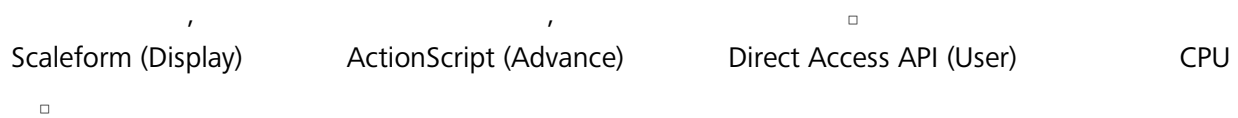


2:AMP

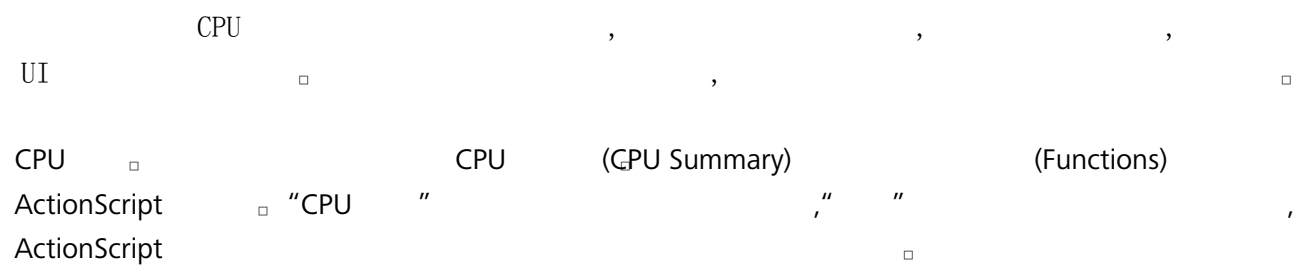
" "(File) " "(Connection) □ AMP
, (PC Xbox360 PS3) □
IP Flash □ AMP
□ Scaleform AMP



2.2



1 CPU

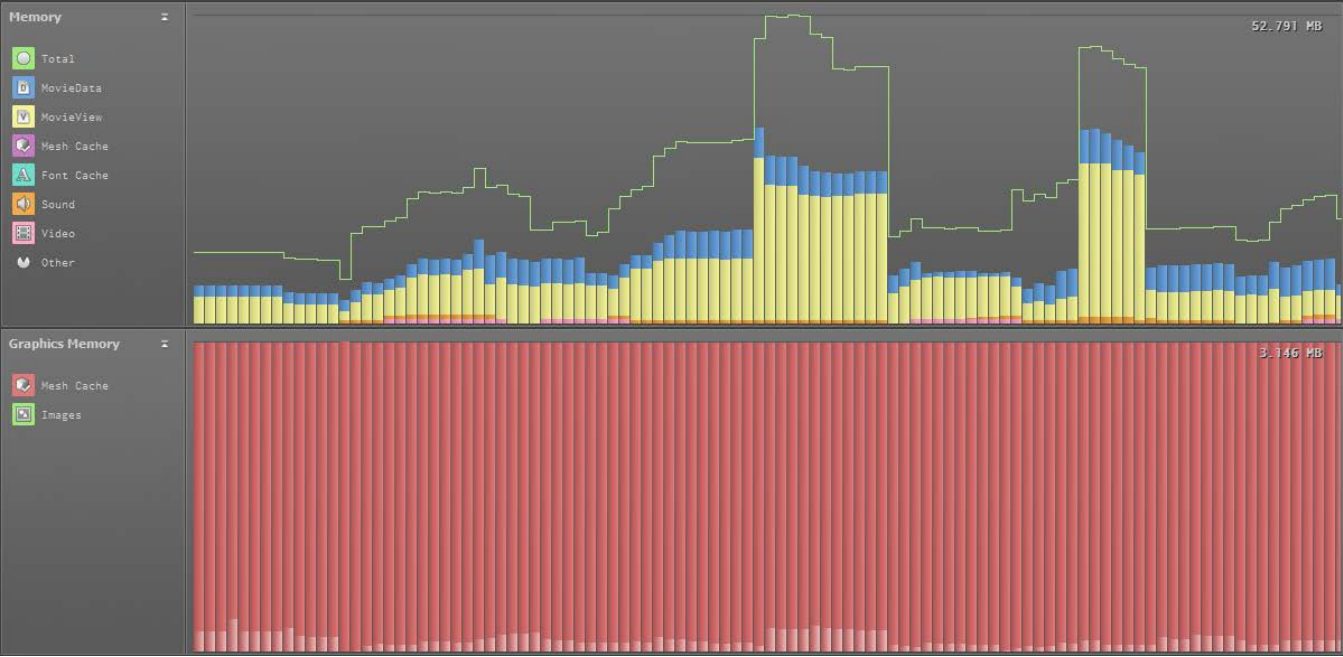


Log	SWF info	Renderer	Memory
Meshes:	182		
Draw Primitives:	183		
Triangles:	5124		
Masks:	6		
Filters:	0		
Strokes:	21		
Gradient Fills:	0		
Rasterized Glyphs:	412		
Font Cache Updates:	0		
Font Textures:	1		
Font Cache Thrashing:	0		
Font Cache Fill:	28%		
Font Cache:	1048576 pixels		
Font Cache Used:	298294 pixels		
Font Failures:	0		
Font Cache Misses:	5		
Mesh Thrashing:	0		

4

2.4

Scaleform :



5

- **(Total)** Scaleform ()
- **(MovieData):** SWF GFX
- **(Movie view):** Gfx::Movie
ActionScript
- **(MeshCache):** (Tesselator)
- **(Font Cache):**
- **(Sound):**
- **(Video):**
- **(Other):** Scaleform
- (Images): ()

[\(Memory System Overview\)](#)

“ ” (Memory)

(),

“Used Space”() GFX

“Unused Space”() GFX

Debug Data() GFX

(AMP i)

Scaleform

Scaleform

- o **Renderer** Scaleform
 - **Buffers** (Mesh Staging Buffer)
 - **RenderBatch** (: ,)
 - **Primitive**
 - **Fill**

- **Mesh**
- **MeshBatch**
- **Context**
- **NodeData**
- **TreeCache**
- **TextureManager**
- **MatrixPool**
- **MatrixPoolHandles**
- **Text**
- **Font**
- o **MovieDef**
- **CharDefs**
- **ShapeData**
- **Tags**
- **Fonts**
- **Sounds**
- **ASBinaryData**
- **MD_Other**
- o **MovieView**
- **MovieClip**
- **ActionScript**
- **ASString**
- **Text**
- **XML**
- **MV_Other**
- **VM**
- **AS3 VM**

- - **CallFrame** ActionScript
 -
 - **Vtable** ActionScript □
 - **SlotInfo** ActionScript □
 - ClassTraits InstanceTraits SlotInfo , (Class)
 - (Instance) □ ClassTraits InstanceTraits ,
 -
 - o **SlotInfoHash** ActionScript
 - (Hash Table) □
 - **ClassTraits** ActionScript □
 - **Class** ActionScript □
 - **InstanceTraits** ActionScript
 -
 - **Instance** ActionScript
 -
 - **AbcFile** **ABC**
 -
 - o **AbcConstPool** **ABC** (
 - Multiname) □ , AbcFile
 -
 - **VMAbcFile** **ABC** ,
 - AbcFile □
 - **Tracer** □
 -
- o **IME** (IME) □

Log	SWF info	Renderer	Memory	Image
Memory			Size	
▲	Total Footprint		4849.402 KB	
▲	Used Space		4633.715 KB	
▷	Video Heaps		0.000 KB	
▷	Other Heaps		325.621 KB	
▲	Movie View Heaps		1226.094 KB	
▲	MovieView "3dgenerator_as3.swf"		1226.094 KB	
	String		0.828 KB	
▷	Renderer		12.672 KB	
▷	MovieView		1176.781 KB	
	General		35.812 KB	
▷	Movie Data Heaps		261.066 KB	
▲	Global Heap		2820.934 KB	
	Video		264.176 KB	
	String		20.633 KB	
	Sound		0.141 KB	
▷	Renderer		2310.914 KB	
▷	MovieView		8.039 KB	
▷	MovieDef		116.676 KB	
	Image		0.141 KB	
	IME		9.914 KB	
	General		90.301 KB	
▷	Debug Memory		8.000 KB	
▷	Unused Space		38.820 KB	
	Debug Data		176.867 KB	

6

()□

Scaleform

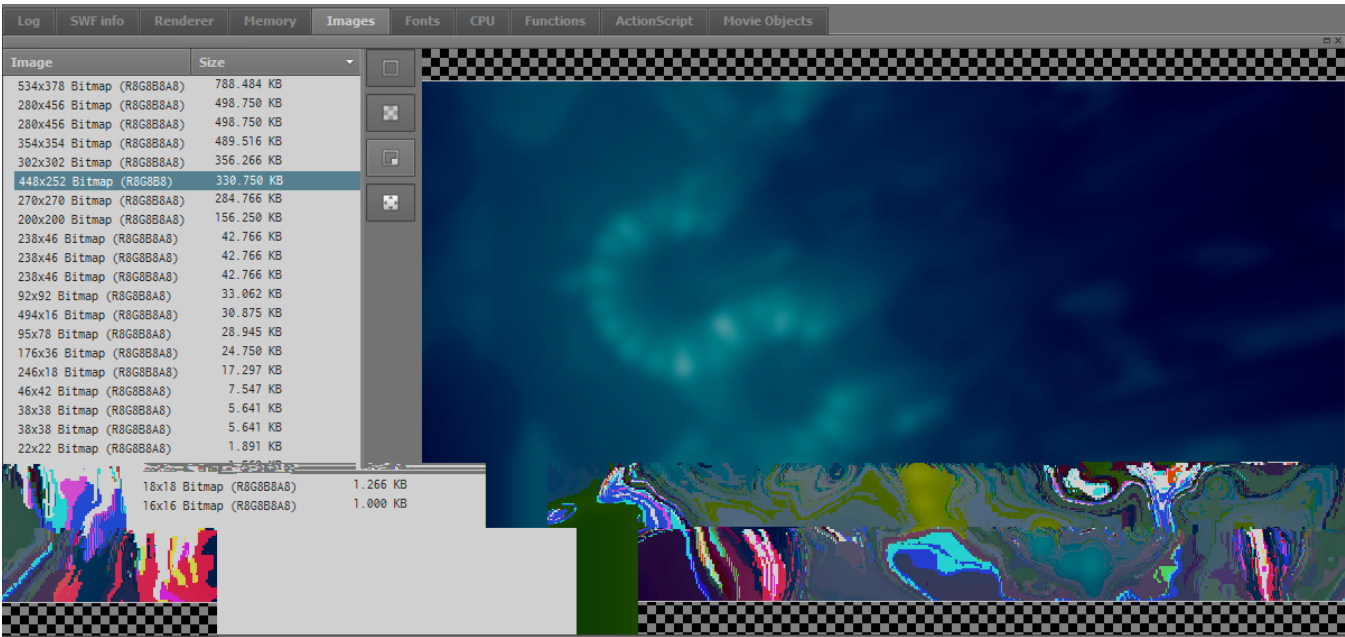
□

“ (Images)

,

,

□



9



Scaleform



Scaleform



(Fonts)

,



10



(SWF)



AMP

MovieData

CharDefs

SWF



" "

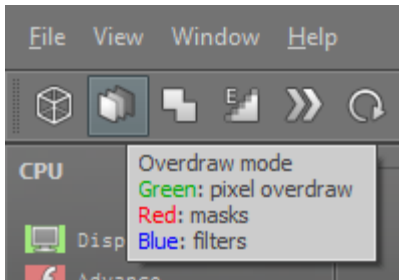
"

"



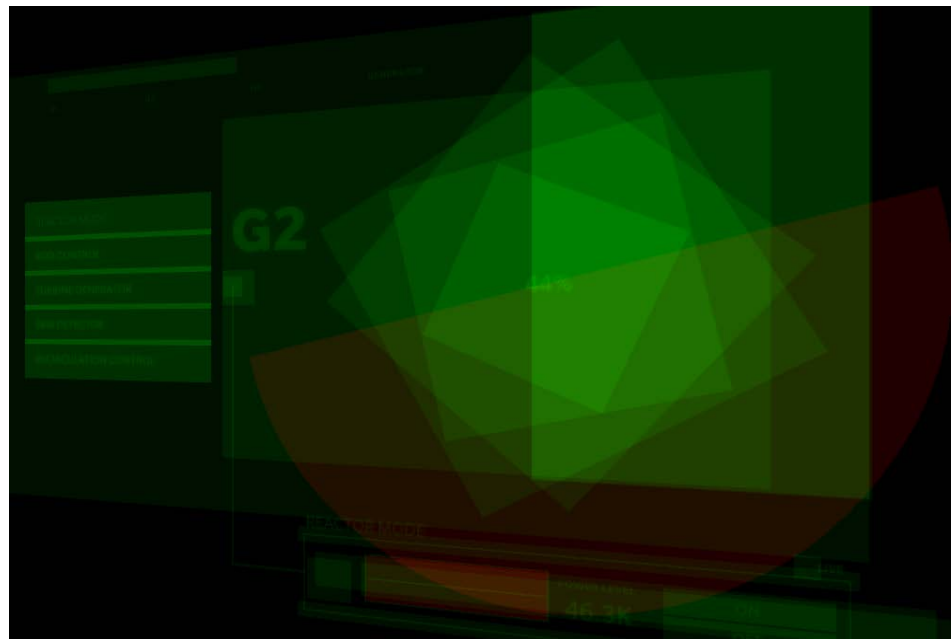
2.5

AMP



11

(8 , (16), ,



12

2.6

□

□

3

1. AMP Gfx::System Gfx::System::Init ,
 Saleform::System Scaleform, Gfx::AMP::Server::Init()
 Gfx::AMP::Server::Uninit() AMP□

1. AMP Windows Xbox360 PS3 Linux Mac, Wii, Android iPhone/iPad, 3DS,
 PS Vita, WiiU□ AMP , □ ,
 AMP IP , , IP □

- Xbox360, IP , IP □ IP Xbox 360
 □

- PS3, IP , IP ,
 IP □

- Android □
 AndroidManifest.xml
 <uses-permission android:name="android.permission.INTERNET" />

- PC (WiFi) WiiU PC 6003
 , TCP/IP

- □ Wii AMP NDEV USB COM
 PC□ IP "wii"
 Wii AMP □ : RVL_SDK/X86/bin □
 AMP hio2.dll

2. (Unreal Engine 3) , AMP
 Flash AS2 SWD/AS3 AS □ , SWD AMP
 , AMP SWF/AS □

4

1. AMP Scaleform AMP,
GfxConfig.h SF_AMP_SERVER, Scaleform Scaleform ,
Ptr<Gfx::AMP::ServerState> serverState = *SF_NEW
Gfx::AMP::ServerState();
serverState->StateFlags &= Gfx::AMP::Amp_Disabled;
AmpServer::GetInstance().UpdateState(serverState);
AMP
2. AMP (Network Socket) ,
(Ws2_32.lib Windows,Xnet.lib Xbox360,libnetctl PS3,
Linux,)
3. AMP , , ,
 , AMP
AMP::Server::GetInstance().SetInitSocketLib(false)
AMP
4. AMP , Scaleform,
AMP::Server::GetInstance().SetHeapLimit()
AMP
5. AMP 7534
AMP::Server::GetInstance().SetListeningPort()
AMP
6. AMP ,
AMP::Server::GetInstance().SetConnectionWaitTime


```

caps.SetRestartMovie(true);
caps.SetToggleAaMode(true);
caps.SetToggleAmpRecording(true);
caps.SetToggleFastForward(true);
caps.SetToggleInstructionProfile(true);
caps.SetToggleOverdraw(true);
caps.SetToggleBatch(true);
caps.SetToggleStrokeType(true);
caps.SetTogglePause(true);
caps.SetToggleWireframe(true);
AMP::Server::GetInstance().SetAppControlCaps(&caps);

```

5.3

```

AMP
AMP::Server::GetInstance().SetConnectedApp
AMP
AMP::Server::GetInstance().SetState
ServerStateType
o Amp_RenderOverdraw
o Amp_App_Wireframe
o Amp_App_Paused
o Amp_App_FastForward
• AMP::Server::GetInstance().SetAaMode
• AMP::Server::GetInstance().SetStrokeType
• AMP::Server::GetInstance().SetCurrentLocale
• AMP::Server::GetInstance().SetCurveTolerance
• AMP::Server::GetInstance().UpdateState
AMP::ServerState
AMP::Server::GetInstance().UpdateState

```

5.4

```

AMP CPU C++ ActionScript
AMP
Amp.addMarker(1)
Amp.addMarker() .as
Resources/AS2/CLIK Amp
C++ Gfx::MovieImpl:: AdvanceStats->AddMarker(1)

```

☐

6 AMP

□

1. AMP

5

1

AMP

1

□

□

(AMP::Server::WaitForAmpConnection)

□

 $\ddot{i}$)

' (profile level)

Gfx::AMP::Server::SetHeapLimit

5.

,AMP 1MB,
 AmpServer::SetHeapLimit
 AMP GfX

7

AMP ,

- [AMP \(AMP Use Cases\)](#)
- [Scaleform Reference](#) – /
- [AMP Forum](#) –